

The Continuation

Semesters 5 through 12 · twenty-four further courses · built after the degree plan was complete

WHAT THIS DOCUMENT IS

The twelve-course MS described in the Program Handbook is a complete degree plan, and Semesters 1 through 4 are that plan. This document records the twenty-four courses built after it.

They are **not part of the degree** and are not required for it. They were written because the curriculum kept pointing at subjects it could not reach in four semesters, and because the cross-references between courses are what make the program read as one thing rather than as thirty-eight.

Same terms as everything else here: **every word written for this program is original and free to copy, adapt, and redistribute**. External courses are linked, never reproduced, and remain under their own licences. The University of Claude is not accredited and awards nothing.

1 How the continuation is organised

Each semester after the fourth is **three courses chosen to converge**. They are not a list of electives: within a semester the three courses arrive at the same conclusion from different directions, and the convergence is stated explicitly in each one's final module.

That structure is the reason to take them in threes rather than individually. **A conclusion reached once is a claim; the same conclusion reached three times — from security, from statistics, and from design — is a result you keep.**

Reading order

- **Semesters 5 through 8 are roughly ordered.** The later ones assume the earlier: Semester 7 uses Semester 6 heavily, and Semester 8 is deliberately placed after the applied work rather than before it.
- **Semesters 9 through 12 are largely independent of each other** and may be taken in any order, though each assumes Semesters 1 through 8.
- **Every course cross-references the others by module and section** — *CSCE nnn Module nn §n* — so a forward reference is a pointer rather than a dependency, and nothing breaks if you read out of order.

2 The eight semesters

Semester 5 — Engine internals

Course	Title
CSCE 620	Computational Geometry
CSCE 605	Compiler Design
CSCE 678	Distributed Systems and Cloud Computing

The three subjects an engine actually runs on once it stops being one program on one machine: **the geometry underneath every spatial structure, the compiler underneath every shader and script, and the distribution underneath every build farm and multiplayer session.**

Semester 6 — Learning foundations

Course	Title
CSCE 633	Machine Learning
CSCE 636	Deep Learning
CSCE 669	Computational Optimization

Learning from data, taken in the order that makes it tractable: **the statistical framing first, the architectures second, and the optimisation that both of them are instances of third.**

Semester 7 — Perception and agents

Course	Title
CSCE 753	Computer Vision
CSCE 625	Artificial Intelligence
CSCE 642	Deep Reinforcement Learning

Systems that take in the world and act in it — **which is the inverse of the rendering problem this track spent four semesters on**, and reads very differently once you have written a renderer.

Semester 8 — Theory

Course	Title
CSCE 627	Theory of Computability
CSCE 637	Complexity Theory
CSCE 658	Randomized Algorithms

What cannot be computed, what can be computed only slowly, and what randomness buys you — **placed after the applied semesters deliberately**, because the negative results mean more once you have tried to build the things they rule out.

Semester 9 — Security

Course	Title
CSCE 701	Foundations of Cybersecurity
CSCE 711	Applied Cryptography
CSCE 713	Software Security

All three written defensively: how systems fail, so that you can build ones that do not. **All three converge on the same conclusion** — make the defect unexpressible rather than training people not to write it: safe defaults, misuse-resistant APIs, and types that make the bug impossible to state.

Semester 10 — Data and language

Course	Title
CSCE 638	Natural Language Processing
CSCE 676	Data Mining
CSCE 670	Information Retrieval

Three subjects that all run on a proxy. **All three converge on the proxy problem** — a metric standing in for quality, a score standing in for a pattern being real, a click standing in for relevance — and on what happens once the proxy becomes the target.

Semester 11 — Human-centred computing

Course	Title
CSCE 671	Human-Computer Interaction
CSCE 679	Data Visualization
CSCE 632	Accessible Computing

All three replace a judgement about taste with a cheap measurement — five participants attempting a real task, the measured ranking of the visual channels, and the actual assistive technology — and **all three correct the same error**: designing for an imagined user who resembles the designer.

Semester 12 — Frontiers

Course	Title
CSCE 640	Quantum Algorithms

Course	Title
CSCE 628	Computational Biology
CSCE 717	Algorithmic Game Theory

In all three the computational model comes from outside computer science and is not yours to choose — physics, biology, and self-interested agents — and **in all three the true state is unobservable**: the amplitudes cannot be read, the ground truth is not available, and the valuations are private.

3 What every course does the same way

The thirty-eight courses share a structure, and it is worth stating plainly because it is what makes them usable without an instructor.

- **Thirteen modules and two project weeks.** Each module is a slide deck and a set of lecture notes, and **the notes say more than the slides do rather than less.**
- **Every module ends in exercises you can actually run,** and a self-check list that is a list of questions rather than a summary.
- **Two projects per course,** one at the midpoint and one at the end, each with its requirements and its done-criteria stated separately — because **the second list is what distinguishes finished from submitted.**
- **Resources are linked, never reproduced,** and each link says which section it is for and why it is worth the time.
- **And every course closes on the same rule,** restated in that course's own vocabulary: *state what you measured, state what you assumed, and never claim more than you established.*

4 The closing rule

That last point is the program's one piece of doctrine, and the final module of every course is an application of it. In CSCE 679 it is a five-clause figure caption. In CSCE 632 it is an accessibility statement with its exceptions named. In CSCE 640 it is the problem, the baseline, and the machine. In CSCE 628 it is the methods section and the artefact account. In CSCE 717 it is the solution concept, the agent model, and the impossibility you chose to live with.

They are the same instruction. A result is a claim about a procedure, and the procedure is the part a reader can check. Stating it is not modesty — it is the difference between work somebody can build on and work they can only believe.

5 Status

38 courses; 483 modules. The twelve-course MS in Semesters 1 through 4, its two prerequisites in Semester 0, and the twenty-four-course continuation in Semesters 5 through 12. Every local link in the repository resolves, and every external link is free to access.

Semesters 13 through 24 are planned and are not built. They appear in the repository index and in the study app as unbuilt, and the catalog maps every one of them to free lectures and books, so any of them can be written later without disturbing what is here. **Nothing in Semesters 0 through 12 depends on them.**

The README at the repository root is generated from the built material and is the current index. **If a course is listed there, its slides and notes exist;** if it is marked as not yet built, they do not.